

Chapter 1: Introduction

1.1 Background

The global maritime shipping industry forms the backbone of international trade, with Very Large Crude Carriers (VLCCs) playing a critical role in the global energy supply chain. Operating a VLCC such as the *Desh Viraat* involves massive daily capital expenditures, the largest of which is bunker fuel. Depending on the transit speed, a VLCC can consume anywhere from 20 to over 70 metric tons of fuel per day.

Simultaneously, the maritime bunker market is characterized by extreme price volatility. Spot prices at major maritime hubs like Fujairah or Singapore fluctuate rapidly based on macroeconomic indicators, geopolitical events, and crude oil indices (such as Brent and WTI). For ship operators and commercial routing desks, deciding how fast to sail and where to purchase fuel are inextricably linked decisions that dictate the profitability of a voyage.

1.2 Problem Statement

Traditional maritime routing and bunkering models suffer from two critical flaws that render them ineffective in real-world commercial operations:

1. **Deterministic Nature :** Legacy models assume deterministic, fixed prices for downstream bunkering hubs. They fail to account for asymmetric market risks (such as a sudden price spike in backwardation or a price drop in contango). Consequently, vessels often over-purchase expensive fuel at the origin port or arrive at a destination port forced to buy at a premium.
2. **The "Eco-Speed" Fallacy:** Basic optimization models attempt to minimize fuel consumption in isolation, which consistently results in the algorithm commanding the vessel to sail at its minimum physical speed (e.g., 8.0 knots). This ignores the immense opportunity cost of time in commercial shipping, quantified as the Time Charter Equivalent (TCE). A vessel crawling across the ocean to save fuel costs the operator hundreds of thousands of dollars in lost chartering days.

There is a critical need for a prescriptive analytics framework that does not merely minimize fuel burn, but maximizes total voyage profitability under market uncertainty.

1.3 Research Objectives

This research aims to bridge the gap between predictive data science and prescriptive operations research by developing a comprehensive decision-support digital twin for VLCC operations. The specific objectives are:

- **Risk Quantification via Machine Learning:** To engineer a predictive engine using LightGBM Quantile Regression that shifts from single-point price forecasting to

asymmetric scenario generation (identifying the 10th, 50th, and 90th percentile price boundaries).

- **Convex Vessel Physics Modeling:** To construct a mathematically rigorous digital twin of a VLCC using the Admiralty Formula, translating raw, non-convex sensor data into a strictly convex cubic relationship between transit speed and fuel consumption, thereby eliminating boundary-condition anomalies in the solver.
- **Stochastic Optimization of Operations:** To formulate and solve a two-stage stochastic programming model that dynamically balances the physical cost of bunker fuel against the Time Charter Equivalent (TCE), outputting precise, actionable directives for optimal speed and staged fuel purchasing.

1.4 Scope and Limitations

The scope of this digital twin and optimization engine is bounded by the following parameters:

- **Vessel Class:** The physical and operational constraints are modeled specifically for a Very Large Crude Carrier (VLCC) profile, calibrated to the specifications of the *Desh Viraat*.
- **Geographical Routing:** The routing matrices and distance calculations are restricted to established crude oil import corridors from the Middle East (e.g., Ras Tanura) to the Indian Subcontinent (e.g., Vadinar, Paradip, Mundra).
- **Market Corridors:** Price modeling is focused on the primary bunkering hubs serving these routes, specifically the Fujairah and Singapore pricing indices.
- **Weather Routing:** This model assumes baseline navigable conditions and does not currently integrate real-time hydrodynamic drag penalties caused by extreme weather systems.

Chapter 2: Predictive Risk Modeling (The ML Engine)

2.1 The Limitation of Deterministic Forecasting

In traditional maritime operations research, fuel prices at downstream bunkering hubs are treated as deterministic inputs—a single expected value based on historical averages or linear extrapolation. However, global bunker markets are highly volatile, driven by macroeconomic shocks, crude oil supply constraints, and geopolitical events.

Standard time-series forecasting models (such as ARIMA) or standard regression models optimize for the Mean Squared Error (MSE). This produces a single-point estimate representing the mean expected price. In a commercial shipping context, optimizing a multi-million-dollar voyage around a mean prediction is financially dangerous. A mean prediction provides no information regarding the *spread* or *asymmetry* of the market risk. If the model predicts a downstream price of \$517 / MT, the commercial desk needs to know if the realistic worst-case scenario is \$525 / MT or \$700 / MT.

To power a stochastic optimization engine, the predictive model must quantify the uncertainty boundaries of the forward market.

2.2 Data Ingestion & Feature Engineering

To capture the underlying market dynamics, the predictive engine utilizes a multi-variate dataset combining direct maritime fuel costs with broader macroeconomic indicators.

- **Target Variable:** The daily spot price of Very Low Sulfur Fuel Oil (VLSFO) at the Fujairah and Singapore bunkering hubs.
- **Macroeconomic Regressors:** Global crude oil indices, specifically Brent Crude and WTI Crude pricing, which serve as leading indicators for refined bunker fuel costs.

To prepare this raw time-series data for the machine learning model, temporal feature engineering was applied to ensure the model could detect momentum and volatility. The dataset was transformed to include rolling averages (e.g., 7-day and 14-day windows), lag features to capture temporal dependencies, and differencing to enforce stationarity in the macroeconomic variables.

2.3 LightGBM Quantile Regression Framework

To solve the problem of single-point deterministic forecasting, this digital twin utilizes **Light Gradient Boosting Machine (LightGBM)** configured specifically for **Quantile Regression**.

LightGBM is a highly efficient, distributed gradient boosting framework that uses tree-based learning algorithms. Unlike standard regression that minimizes the L2 loss (Mean Squared Error), quantile regression minimizes the **Pinball Loss** (or Quantile Loss) function. This allows the model to predict specific percentiles of the target distribution.

The Pinball Loss function $L_\tau(y, \hat{y})$ for a given quantile τ is mathematically defined as:

$$\mathcal{L}_\tau(y, \hat{y}) = \begin{cases} \tau |y - \hat{y}|, & \text{if } y \geq \hat{y}, \\ (1 - \tau) |y - \hat{y}|, & \text{if } y < \hat{y}. \end{cases}$$

Where:

- y is the actual realized downstream bunker price.
- $\hat{y}$ the predicted bunker price.
- τ is the target quantile (ranging from 0.0 to 1.0).

By penalizing over-predictions and under-predictions asymmetrically based on the value of τ , the model can be trained simultaneously on multiple quantiles to construct a probabilistic boundary of future prices.

2.4 Asymmetric Scenario Generation

For the optimization engine to make actionable hedging decisions, the continuous probabilistic output of the LightGBM model was discretized into three distinct forward market scenarios.

Three separate LightGBM regressors were trained with τ values of 0.10, 0.50, and 0.90.

These outputs translate directly into the discrete scenarios injected into the Stage-2 Gurobi solver:

- **Scenario S_Low (10th Percentile):** Represents a market drop . Assigned a realization probability of 20 %.
- **Scenario S_Base (50th Percentile):** Represents the median expected market condition. Assigned a realization probability of 50 %.
- **Scenario S_High (90th Percentile):** Represents a severe market spike (Backwardation). Assigned a realization probability of 30 %.

By engineering the machine learning model to output this specific asymmetric risk matrix, the downstream operations research algorithm is empowered to mathematically "hedge" its bunker purchases—buying just enough fuel to survive if a price drop is highly probable, or filling the tanks to maximum capacity to insulate the vessel against an impending price spike.

Chapter 3: Vessel Physics & The Digital Twin

3.1 Raw Data Limitations and the Non-Convexity Problem

In the initial stages of digital twin development, the fuel consumption model for the *Desh Viraat* was constructed using raw, empirical sensor data extracted directly from ship logs. While empirical data is theoretically highly accurate, raw sensor readings inherently contain noise caused by varying drafts, bio-fouling, and ocean currents.

When this raw data was fed into the optimization engine, the marginal fuel cost per knot exhibited non-convex behavior. Specifically, the data showed that accelerating from 11.6 to 12.8 knots required less marginal fuel than accelerating from 10.4 to 11.6 knots.

Linear and mixed-integer solvers (like Gurobi) are ruthless mathematical engines. When presented with a non-convex performance curve featuring a flattened upper bound, the solver exploits it. The algorithm determined that if the vessel was already sailing at an interior speed, the marginal penalty to push to maximum speed was economically justified. This resulted in a "Bang-Bang" boundary solution, where the optimization engine completely ignored interior speeds (e.g., 11.5 knots) and strictly alternated between the physical minimum (8.0 knots) and the physical maximum (14.0 knots). To create a commercially viable model, the physics engine required mathematical smoothing.

3.2 The Admiralty Formula and Strict Convexity

To resolve the boundary-solution anomaly, the empirical data was replaced with a strictly convex theoretical model based on the **Admiralty Formula**. The Admiralty coefficient is a fundamental principle in naval architecture which dictates that the propulsive power required to overcome hydrodynamic drag—and consequently, the fuel consumption—is proportional to the cube of the vessel's speed.

$$C = \frac{D^{\frac{2}{3}} \cdot V^3}{P}$$

Where,

C = Admiralty Constant , D = Displacement of the Ship, V = Speed of the Vessel, P = Shaft Power

By enforcing a cubic relationship, the model guarantees strict convexity. As the vessel accelerates, the hydrodynamic resistance increases exponentially, forcing the optimization engine to accurately weigh the escalating cost of fuel against the time saved.

The daily fuel consumption function $F(v)$ was modeled as:

$$F(v) = B + c * v^3$$

Where:

- $F(v)$ is the total daily fuel consumption in Metric Tons (MT).
- B is the baseline auxiliary load (fuel required for onboard generators and systems independent of propulsion).
- c is the hull resistance constant specific to the Vessel *Desh Viraat*.
- v is the transit speed in knots.

3.3 Deriving the Vessel Coefficients -

To calibrate the Admiralty formula specifically for the *Desh Viraat*, the theoretical curve was anchored to the vessel's validated extreme performance points:

- Minimum Speed : At 8.0 knots, total consumption is 17.62 MT/day.
- Maximum Speed : At 14.0 knots, total consumption is 51.02 MT/day.

Solving the system of equations derived from these anchors yielded the following physical constants for the digital twin:

- **Auxiliary Base Load (B):** 9.96 MT/day
- **Hull (Admiralty) Constant (c):** 0.01496

Replacing the raw six-point data table with a high-resolution, 25-point cubic interpolation allowed the spatial branch-and-bound solver to discover highly specific, interior optimal speeds (e.g., 11.4 knots) that balance the Time Charter Equivalent against fuel burn.

3.4 Operational and Volumetric Constraints

Beyond speed and consumption, the digital twin must operate within the strict physical safety margins of a maritime vessel. The following constraints were hard-coded into the vessel profile:

1. **Tank Capacity and Density:** The absolute volumetric limit of the bunker tanks (7,845.5 m³) was converted to a mass limit using the specific gravity of VLSFO (0.96 MT/m³) and a maximum safe fill factor of 0.98, establishing a hard upper ceiling for any Stage-1 or Stage-2 purchasing decision.
2. **Safety Reserves:** To account for unpredictable weather or port delays, the model enforces a strict Minimum Safety Reserve constraint of 500.0 MT. The optimization engine is mathematically forbidden from calculating a route where the projected fuel onboard drops below this threshold.
3. **Deadweight Draft Penalty:** Because carrying excess fuel increases the vessel's weight and draft, dragging it deeper into the water, a dynamic consumption multiplier was integrated. For every 1,000 MT of bunker fuel onboard, the baseline fuel consumption is penalized by 0.5 %. This forces the solver to recognize the hidden cost of "tankering" (carrying excess fuel to avoid buying at expensive ports).

Chapter 4: Mathematical Formulation

4.1 Model Architecture: Two-Stage Stochastic Programming

The core of the decision-support system is built upon a Two-Stage Stochastic Programming architecture. In maritime logistics, decisions cannot be made in a vacuum; they must account for future uncertainties. A standard deterministic model assumes the future is known perfectly, which leads to fragile, suboptimal strategies in volatile bunker markets.

The stochastic model divides the decision-making timeline into two distinct stages:

- **Stage 1 (The "Here and Now" Decision):** The algorithm must determine the immediate bunker purchase quantity at the origin port (e.g. Ras Tanura) based on the known current spot price, *before* the downstream market fluctuations reveal themselves.
- **Stage 2 (The Recourse Decisions):** These are the operational adaptations the vessel makes once the future market state becomes known. For each discrete scenario generated by the LightGBM engine (S_Low, S_Base, S_High), the solver calculates a specific recourse strategy. This includes the optimal transit speed and the downstream bunker purchase required at the destination hub.

4.2 The Objective Function: Profitability Maximization

Historically, maritime optimization heavily focused on the heuristic of "Eco-Speed" minimizing physical fuel consumption. However, minimizing fuel burn inherently maximizes voyage duration. This ignores the commercial value of the vessel, represented by the Time Charter Equivalent (TCE).

The objective of this digital twin is not to minimize fuel, but to minimize the **Total Expected Voyage Cost**, which dynamically balances the capital expenditure of fuel against the opportunity cost of time.

The objective function is formulated to minimize \$Z\$:

$$\min Z = C_{\text{spot}} Q_1 + \sum_{s \in \mathcal{S}} P_s (C_s Q_{2,s} + \text{TCE} \cdot T_s)$$

Where:

- Q_1 = Bunker quantity purchased at Stage 1 (Origin).
- C_{spot} = Current spot price of fuel at Origin.
- $\mathcal{S}$ = Set of future market scenarios $\{\mathcal{S}_{\text{Low}}, \mathcal{S}_{\text{Base}}, \mathcal{S}_{\text{High}}\}$.
- P_s = Probability of scenario s occurring.
- C_s = Forecasted bunker price at the downstream hub under scenario s .
- $Q_{2,s}$ = Recourse bunker quantity purchased at Stage 2 under scenario s .
- TCE = Daily Time Charter Equivalent rate (Opportunity Cost).
- T_s = Total voyage duration in days under scenario s , defined as $T_s = D / (24 \cdot v_s)$ (where D is total distance and v_s is the vessel speed).

4.3 Constraints Definition

To ensure the mathematical outputs are physically executable by the vessel's crew, the solver must operate within strict bounds. During development, a critical logical hazard was identified where the solver would theoretically purchase cheaper Stage-2 fuel to retroactively satisfy Stage-1 consumption (a flow-balance violation).

To enforce the laws of physics and maritime safety, the following constraints were strictly defined for all scenarios s in $\{\mathcal{S}\}$:

1. Forward Flow Balance :

The fuel currently onboard (I_{init}) plus the fuel purchased at Stage 1 (Q_1) must be sufficient to complete the voyage burn ($F_{\{\text{burn}, s\}}$) and still satisfy the minimum safety reserve (R_{min}) upon arrival, *before* any Stage-2 fuel can be loaded.

$$I_{\text{init}} + Q_1 - F_{\text{burn},s} + Q_{2,s} \leq \text{Cap}_{\text{max}}, \quad \forall s \in \mathcal{S}$$

(Where $F_{\text{burn},s} = T_s * (B + c * v_s^3) * (\text{DraftMultiplier})$)

2. Volumetric Capacity Constraint:

The total fuel in the tanks immediately following the Stage-2 purchase cannot exceed the physical capacity (Cap_{max}) of the *Desh Viraat*.

$$I_{\text{init}} + Q_1 - F_{\text{burn},s} + Q_{2,s} \leq \text{Cap}_{\text{max}}, \quad \forall s \in \mathcal{S}$$

3. Boundary Speed Constraints:

The transit speed selected for any scenario must fall within the physical operating limits of the vessel's engine.

$$8.0 \leq v_s \leq 14.0, \quad \forall s \in \mathcal{S}$$

4. Non-Negativity Constraints:

Purchasing decisions cannot be negative (the vessel cannot sell fuel back to the port).

$$Q_1 \geq 0, \quad Q_{2,s} \geq 0, \quad \forall s \in \mathcal{S}$$

4.4 Spatial Branch-and-Bound and Non-Convexity

Because the objective function incorporates voyage time ($T_s = D / (24 * v_s)$) and the constraints incorporate the Admiralty cubic physics (v_s^3), the resulting mathematical space features non-linear multiplicative terms.

Standard linear programming (LP) solvers cannot resolve these interacting non-linear decision variables. Therefore, utilizing the **Gurobi Optimizer** for its advanced Spatial Branch-and-Bound algorithm (enabled via the NonConvex=2 parameter). This allows the solver to partition the continuous, non-linear search space into smaller, manageable convex regions, guaranteeing the discovery of the global mathematical minimum for the optimal speed and bunkering strategy rather than getting trapped in a local optimum.

Chapter 5: Results & Economic Inferences

5.1 Experimental Setup and Computational Environment

Before analyzing the economic outcomes, it is necessary to establish the computational environment that generated them. The Two-Stage Stochastic model was executed using the Gurobi Optimizer via its Python API. Because the objective function and physics constraints

introduced non-linear, multiplicative decision variables (specifically the inverse relationship of time to speed, $T_s = D / 24 v_s$, combined with cubic fuel burn, v_s^3), the problem space was non-convex.

The Gurobi parameter NonConvex=2 was initialized, forcing the engine to utilize the spatial branch-and-bound algorithm. This ensures the solver searches for the global mathematical minimum rather than terminating at a local optimum. The computational latency for solving a complete three-scenario recourse matrix across a 3,450 NM voyage (Route R05) averaged under 2.5 seconds, proving the model's viability for real-time commercial desk deployment.

5.2 Validation of the Digital Twin Physics

The first phase of the results focuses on the successful mitigation of the boundary-condition anomaly (the "Bang-Bang" effect). In earlier iterations using raw sensor data, the linear solver repeatedly bounded the vessel speed to either 8.0 knots or 14.0 knots due to the flattened upper limit of the empirical fuel curve.

Upon implementing the Admiralty Formula to enforce strict convexity ,

$$(F(v) = 9.96 + 0.01496 * v^3)$$

the spatial branch-and-bound solver successfully identified interior optimal speeds. For example, under a Base Market scenario with a TCE of \$40,000 / day and a fuel price of \$517 / MT, the optimal speed was calculated precisely at 11.4 knots. This validates that the mathematical engine is perfectly balancing the exponentially increasing physical drag against the linear accumulation of time-based opportunity costs.

5.3 The Mathematical Independence of Speed and Voyage Distance

A major operational insight derived from this model counters traditional maritime intuition: the optimal transit speed is mathematically independent of the total voyage distance.

The optimization matrices were executed across three distinct crude import corridors:

- Route R01: Ras Tanura to Vadinar (1,650 NM)
- Route R02: Ras Tanura to unassigned hub (2,500 NM)
- Route R05: Ras Tanura to Paradip (3,450 NM)

Despite the massive variance in total physical fuel required for these routes, the solver outputted the exact same optimal speed for a given TCE across all three distances. By factoring distance (D) out of the total cost derivative, the model proves that optimal speed is governed strictly by the ratio between the fuel price, the hull resistance constant, and the TCE. Distance acts solely as a scalar multiplier on the total cost, but it does not shift the coordinate of the minimum cost on the efficiency frontier.

5.4 The Bunkering Arbitrage Strategy (Stage-1 vs. Stage-2)

The integration of the LightGBM asymmetric price forecasts allowed the Gurobi solver to execute highly sophisticated hedging strategies, acting effectively as an algorithmic trading desk. The Stage-1 immediate purchase decision (Q_1) reacted dynamically to the downstream predictions:

- **Contango Mitigation (Downstream is Cheaper):** When the ML model predicted a high probability of a price drop at the destination hub, the solver refused to fill the tanks at Ras Tanura. Instead, it purchased exactly 350 MT, the absolute minimum required to physically survive the Stage-1 transit burn plus the mandatory 500 MT safety reserve. The bulk of the procurement was mathematically delayed to Stage 2.
- **Backwardation Hedging (Downstream Price Spike):** When the 90th percentile LightGBM scenario triggered an expected price spike, the solver aggressively capitalized on the current spot price, maxing out the tank capacity at 784 MT in Stage 1 to insulate the voyage from the impending downstream premium.

5.5 The Time Charter Equivalent (TCE) Shift Effect

The economic impact of the opportunity cost was visualized by mapping the continuous optimization frontier across varying market climates. The total cost function (Z) is U-shaped. As the TCE artificially escalates the penalty for time spent at sea, the entire U-curve shifts downward and to the right.

- Weakened Market (\$ 20,000/day): The dominant cost is physical fuel. The mathematical minimum sits far to the left, resulting in an optimal speed of 8.0 knots.
- Booming Market (\$ 80,000/day): The dominant cost is lost time. The mathematical minimum shifts drastically to the right. The engine demands maximum speed (14.0 knots), willingly absorbing a massive increase in physical bunker spend because the time saved frees up the asset for the next lucrative charter contract.

5.6 Impact of Volumetric and Draft Penalties

The model successfully captured the hidden costs of "tankering" fuel. By implementing a 0.5 % consumption penalty per 1,000 MT of fuel weight, the solver actively avoided purchasing excessive fuel unless the downstream price spread was wide enough to justify the penalty. In flat markets where the spot price and the downstream predicted price were roughly equal, the solver defaulted to smaller, just-in-time Stage-1 purchases to keep the vessel riding higher in the water, thereby minimizing hydrodynamic drag and reducing overall operational expenditure.

5.7 The Coupled Constraint: Speed vs. Capital Outlay

A counter-intuitive interaction was discovered between the market rate (TCE) and the immediate Stage-1 capital outlay. In a boom market (\$80k TCE), the ship sails faster to save time. However, due to the cubic physics model, this faster speed burns exponentially more fuel.

Consequently, to guarantee the vessel arrives with the mandatory 500 MT safety reserve intact, the solver is forced to authorize a much larger Stage-1 fuel purchase purely to cover the aggressive burn rate. This proves a critical coupling effect: optimizing for time directly dictates the required liquidity and capital outlay at the origin port.

5.8 Summary of Economic Viability

By shifting the objective function from pure fuel minimization to a holistic stochastic profit maximization, this digital twin prevents commercial routing desks from making myopic decisions. It proves that slowing a ship down to save \$50,000 in fuel is mathematically incorrect if it costs the company \$150,000 in lost chartering time. The Two-Stage Stochastic framework provides a mathematically provable, executable blueprint for VLCC operations under deep market uncertainty.

Chapter 6: Conclusion & Future Scope

6.1 Summary of the Study

This research proposed and validated a Two-Stage Stochastic Programming framework for optimizing the bunkering strategy and transit speed of a Very Large Crude Carrier (VLCC). By integrating predictive data science with prescriptive operations research, the study developed a decision-support model capable of operating under bunker market uncertainty. LightGBM Quantile Regression was utilized to generate asymmetric forward-price scenarios, which were subsequently fed into a spatial branch-and-bound optimization engine (Gurobi). The model successfully shifted the operational objective from isolated fuel minimization to holistic profit maximization by incorporating the Time Charter Equivalent (TCE) as a foundational constraint.

6.2 Key Findings and Operational Inferences

The experimental results yielded several critical insights regarding maritime logistics and vessel optimization:

- **Mitigation of Non-Convexity:** The application of the Admiralty Formula mathematically smoothed the empirical sensor data into a strictly convex cubic function. This intervention successfully eliminated the boundary-speed anomaly ("Bang-Bang" control), allowing the solver to identify precise, interior optimal transit speeds.
- **Independence of Distance and Speed:** The mathematical derivation of the total cost function, corroborated by the computational outputs across multiple routes (R01, R02,

R05), demonstrated that optimal transit speed is independent of voyage distance. It is dictated entirely by the ratio of current bunker prices to the TCE.

- **Stochastic Hedging Behavior:** The two-stage architecture demonstrated the capacity to algorithmically hedge against price volatility. In simulated contango markets, the model restricted Stage-1 purchases to the physical survival minimum (\$350\text{ MT}\\$). Conversely, in anticipated backwardation markets, the model maximized Stage-1 tank utilization to insulate the voyage against downstream premiums.
- **The Coupled Capital Constraint:** The study revealed a direct correlation between market strength and initial capital outlay. High TCE environments necessitated faster transit speeds, which exponentially increased fuel burn, thereby mandating larger initial bunker purchases simply to maintain baseline safety reserves.

6.3 Limitations of the Current Model

While the proposed digital twin provides a mathematically robust framework, its application is currently constrained by specific deterministic assumptions:

- **Static Environmental Variables:** The model operates under the assumption of baseline navigable conditions. The hull resistance constant (c_h) remains static throughout the voyage, failing to account for dynamic hydrodynamic drag induced by real-time variations in significant wave height, ocean currents, and wind resistance.
- **Single-Node Recourse:** The current routing matrix assumes a single, predetermined downstream discharge and bunkering hub. It does not account for the operational flexibility required to divert the vessel to alternative geographical hubs based on emerging price spreads during transit.

6.4 Avenues for Future Research

To further enhance the commercial viability and operational accuracy of this digital twin, the following extensions are proposed for future study:

- **Integration of Hydrodynamic Weather Routing:** Future iterations of the model should couple the stochastic economic engine with a real-time meteorological API. Transforming the hull resistance constant into a dynamic time-series variable, $c_h(t)$, would allow the optimization engine to simultaneously solve for both the optimal speed and the optimal geographic path to evade severe weather systems.
- **Multi-Echelon Network Optimization:** The Stage-2 recourse matrix can be expanded into a multi-node network graph. Formulating the problem to allow the solver to select from a set of downstream hubs (e.g., Fujairah, Singapore, Colombo) would exponentially increase the arbitrage capabilities of the model.
- **Real-Time Deployment Architecture:** To bridge the gap between theoretical operations research and practical industrial application, the underlying optimization pipeline should

be deployed as a web-based interactive framework. This would allow commercial routing desks to input live market variables and immediately extract optimal routing directives.